

Experiment 1

Code :

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

data = pd.read_csv("dataset.csv")

data.columns = data.columns.str.strip()

print("Columns in dataset:", data.columns.tolist())

if "Age" not in data.columns or "Glucose" not in data.columns:

    print("\nError: Required columns 'Age' and 'Glucose' were not found.")

    print("Available columns are:", data.columns.tolist())

    exit()

age = data["Age"].to_numpy()

glucose = data["Glucose"].to_numpy()

plt.figure(figsize=(8,6))

plt.scatter(age, glucose, color='blue', s=70, label='Original Data')

plt.xlabel("Age (X)")

plt.ylabel("Glucose Level (Y)")

plt.title("Age vs Glucose Level")

plt.grid(True)

plt.legend()

plt.show()

mean_x = np.mean(age)

mean_y = np.mean(glucose)

b1 = np.sum((age - mean_x) * (glucose - mean_y)) / np.sum((age - mean_x) ** 2)

b0 = mean_y - b1 * mean_x

print("\nRegression Coefficients")

print("-----")

print("Intercept (b0):", round(b0, 4))

print("Slope (b1):", round(b1, 4))

print("-----")

print(f"Y = {b0:.4f} + ({b1:.4f})X")

predicted = b0 + b1 * age

sorted_index = np.argsort(age)

age_sorted = age[sorted_index]

predicted_sorted = predicted[sorted_index]

plt.figure(figsize=(8,6))
```

```

plt.scatter(age, glucose, color='blue', s=70, label='Original Data')

plt.plot(age_sorted, predicted_sorted, color='red', linewidth=2, label='Regression Line')

plt.xlabel("Age (X)")

plt.ylabel("Glucose Level (Y)")

plt.title("Linear Regression: Age vs Glucose Level")

plt.grid(True)

plt.legend()

plt.show()

print("\nInterpretation")

print("-----")

if b1 > 0:

    print("The slope is positive.")

    print("As age increases, glucose level tends to increase.")

elif b1 < 0:

    print("The slope is negative.")

    print("As age increases, glucose level tends to decrease.")

else:

    print("No linear relationship exists between age and glucose level.")

new_age = 55

predicted_glucose = b0 + b1 * new_age

print("\nPrediction")

print("-----")

print(f"Predicted Glucose Level for Age {new_age} = {predicted_glucose:.2f}")

```

Output:

Columns in dataset: ['Age', 'Glucose']

Regression Coefficients

Intercept (b0): 65.1416

Slope (b1): 0.3852

$Y = 65.1416 + (0.3852)X$

Interpretation

The slope is positive.

As age increases, glucose level tends to increase.

Prediction

Predicted Glucose Level for Age 55 = 86.33

